

FLAT DISPLAYS

The Future Is Nano

Nano-emissive displays are potentially the cheapest and best alternative to HD TV, CRT and LCD screens. These can be made in larger sizes with better image quality at a lesser cost

■ DR S.S. VERMA

Cathode ray tube (CRT) is a cheap, versatile and high-quality imaging device that has governed the display technology for years. However, with technological developments, other forms of displays like liquid-crystal displays (LCDs), plasma displays and HDTVs have emerged that are said to be better than CRTs.

There are various drawbacks associated with CRTs. These are bulky and voltage-hungry, have limited screen regulation, and emit electromagnetic radiations. LCDs and plasma displays are costlier but offer the advantages of a flat screen, thin size, light weight and more screen regulation. Also, these can hang on just about any surface with appropriate mounting equipment, and are travel-friendly and easy to use for tailgating.

As high-definition DVD formats, television broadcasts and next-generation video games attract consumers, manufacturers are looking for ways to reduce the production costs of HD screens and make them affordable for consumers. LCD requires a very complex manufacturing process and HDTV displays are more costly to produce. Besides, these technologies have other limitations like limited viewing angle. So manufacturers of electronic displays are beginning to explore cheaper alternatives to LCD and plasma screens.

With the latest developments in nano technology, new versions of



Nano-emissive display by Motorola

emissive displays called nano emissive displays (NEDs) are expected to revolutionise small- and big-scale displays with many advantages to the customers.

FEDs combine CRT and LCD advantages

Conventionally, a CRT is powered by an electron gun in which electrons are created by heating a metal filament (process known as 'thermionic emission'). The electrons are then accelerated and focused into a fast-moving beam onto the screen through surrounding electromagnets, allowing it to be scanned across the screen to produce a 2-D display.

A field-emission display (FED) is a flat-panel display technology that uses large-area field electron sources to provide electrons that strike coloured phosphor to produce a colour image. In a general sense, an FED consists of a matrix of cathode ray tubes, each tube producing a single sub-pixel, grouped in three to form red-green-blue (RGB) pixels.

FEDs combine the advantages of CRTs with the packaging advantages of LCD and other flat-panel technologies. These also offer the possibility of lesser power consumption—about half that of an LCD system.

FEDs create light directly on the screen in a fashion identical to a CRT,